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Total No. of Questions : 05]

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ZE-196

B.Tech. (BE) (New Scheme) (Main/ATKT)

Examination, 2026

(Fourth Semester)

MICROPROCESSOR AND

MICROCONTROLLER

EL-405

Time : 3 Hours]

[Maximum Marks : 60

Note : All questions are compulsory. All questions carry equal marks.

1. (a) Explain instruction set of 8085 microprocessor.
- (b) Explain the interrupts of 8085 microprocessor.

Or

- (a) Discuss in detail Data Transfer Scheme in 8085 microprocessor.

- (b) Explain memory interfacing in 8085 microprocessor.
2. (a) Draw and discuss the internal architecture of 8086.
- (b) Explain real and protected mode of operation used in memory management.

Or

- (a) Explain basic organization of 8086 microprocessor.
- (b) Discuss real and protected mode in a 8086 microprocessor.
3. (a) Explain the advancement in Intel series processor (8086 onwards).
- (b) Discuss in detail UltraSPARC processor.

Or

- (a) Explain RISC processor.
- (b) Explain super scalar architecture of Pentium.

4. (a) What do you mean by interfacing in microprocessor ?
(b) Write a short note on IC 8251.

Or

- (a) Draw the block diagram of IC 8259 and explain its operation.
(b) Draw and explain block diagram of 8257 DMA controller.
5. (a) Draw and discuss the internal architecture of IC 8051.
(b) Discuss in detail the instruction set of 8051 microcontroller.

Or

- (a) Explain pin description of 8051 microcontroller.
(b) Explain addressing modes of 8051 microcontroller.

